American Computer Science League

2023-2024 ● Contest 4: Short Problems ● Elementary Division

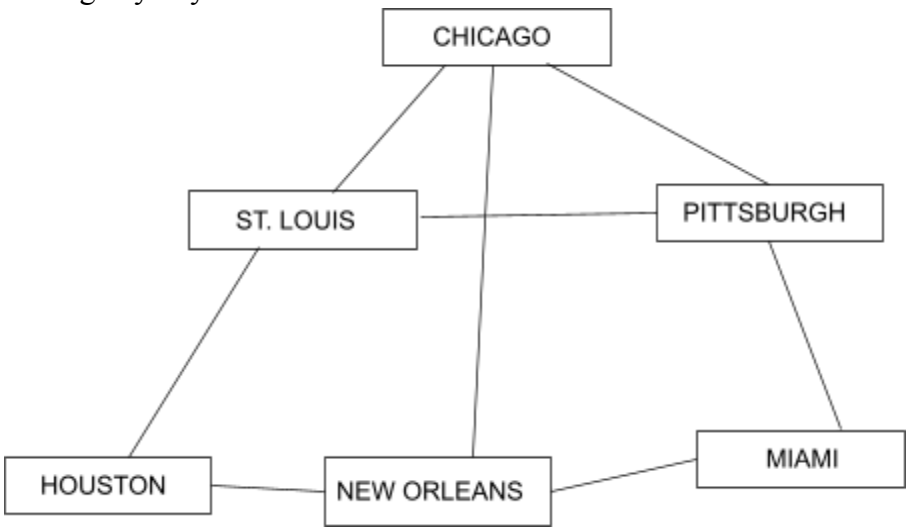
1. Graph Theory

If you draw all sides and diagonals in a regular polygon, you will end up with a complete graph. How many diagonals are there in an octagon?

A. 14
B. 19
C. 20
D. 28
E. 36

2. Graph Theory

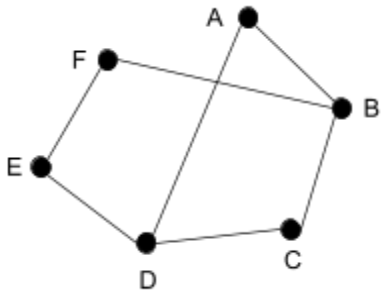
Given the following graph that represents various airplane routes between cities, how many different ways can you get from Chicago to Miami without visiting any city twice?



A. 4
B. 5
C. 6
D. 7
E. 8

3. Graph Theory

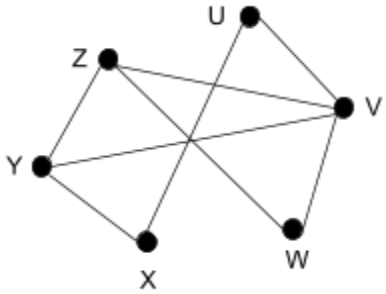
The following graph represents six friends in a school texting each other individually. Their names are Ann, Beth, Carol, Dawn, Eve, and Fran. If Ann is sending a message using exactly 2 in-between friends, which friend is unable to receive the text?



A. B
B. C
C. D
D. E
E. F

4. Graph Theory

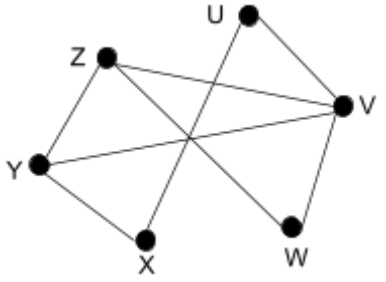
Which of the following is a valid traversal of the graph below? (This is the same graph as #5.)



A. YVUXYVWVZ
B. YZWVUXYVZ
C. YZWVUXYZV
D. YXUVZWVZY
E. All of the above

5. Graph Theory

How many cycles are in the following graph? (This is the same graph as #4.)



A. 6
B. 8
C. 10
D. 12
E. 14

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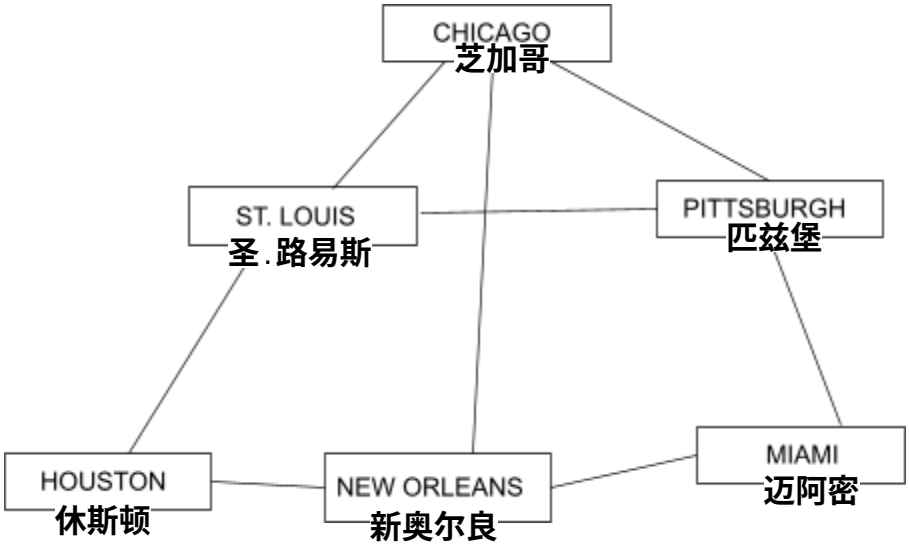
1. 图论

如果**你画出一个正多边形所有的边和对角线，最终会得到一个**完全图。**请问一个八边形有多少条对角线？**

A. 14
B. 19
C. 20
D. 28
E. 36

2. 图论

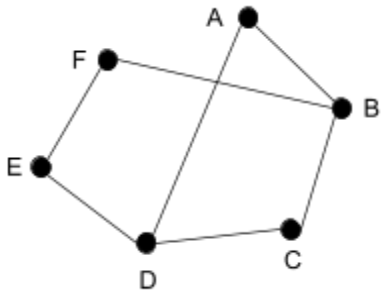
已知**下图为各城际航线图，请问有多少种方式能够让你从芝加哥飞往迈阿密**且不用在某一**城市辗转两次？**



A. 4
B. 5
C. 6
D. 7
E. 8

3. 图论

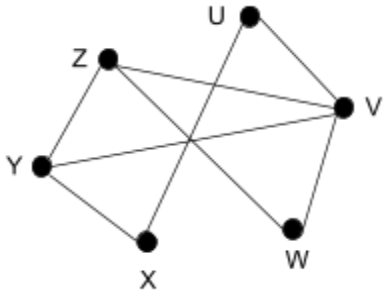
下图**表示同一个学校的六位朋友互相单独给对方发短信。这六位朋友分别是** Ann、Beth、Carol、Dawn、Eve、Fran 。**如果** Ann **恰好通过** 2 **位**中间**朋友**来**发短信，那么请问哪位朋友无法收到短信？**



A. B
B. C
C. D
D. E
E. F

4. 图论

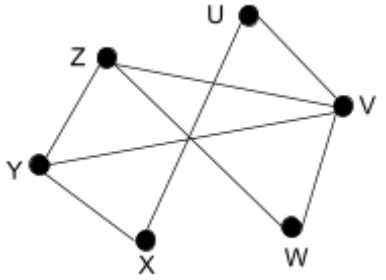
请问**哪项可以有效遍历下图？**(下图**与第** 5 **题图相同。**)



A. YVUXYVWVZ
B. YZWVUXYVZ
C. YZWVUXYZV
D. YXUVZWVZY
E. 以上全部

5. 图论

请问**下图中有多少个环？**(下图**与第** 4 **题图相同。**)



A. 6
B. 8
C. 10
D. 12
E. 14